In favor of weird, exploratory vegan research

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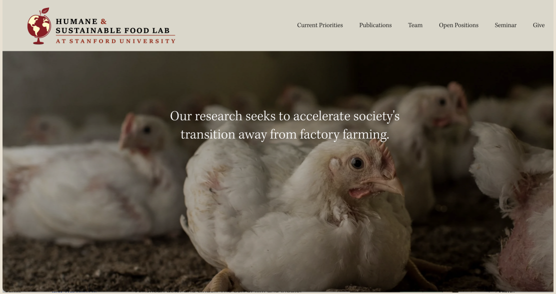
HSFL overview



- Hi, I'm Seth, I'm a Research scientist at the Humane & Sustainable Food Lab at Stanford (foodlabstanford.com)
- The lab was started in early 2023 to accelerate the end of factory farming
- I joined July 2024, I write meta-analyses (and a Substack) and run experiments

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Meaningfully reducing MAP consumption is hard

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Full Length Article

Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis

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A B S T R A C T

Which interventions produce the largest and most enduring reductions in consumption of meat and animal products (MAP)? We address this question with a theoretical review and meta-analysis of randomized controlled trials that measured MAP consumption at least one day after intervention. We meta-analyze 35 papers comprising 41 studies, 112 interventions, and approximately 87,000 subjects. We find that these papers employ four major strategies to change behavior: choice architecture, persuasion, psychology (manipulating the interpersonal, cognitive, or affective factors associated with eating MAP), and a combination of persuasion and psychology. The pooled effect of all 112 interventions on MAP consumption is quite small (standardized mean difference (SMD) = 0.07 (95 % CI: [0.02, 0.12]), indicating an unsolved problem. Interventions aiming to reduce only consumption of red and processed meat were more effective (SMD = 0.25; 95 % CI: [0.11, 0.38]), but it remains unclear whether such interventions also decrease consumption of other forms of MAP. We conclude that while existing approaches do not provide a proven remedy to MAP consumption, designs and measurement strategies have generally been improving over time, and many promising interventions await rigorous evaluation.

- My firs project was a Meta-analytic review of meat and animal product (MAP) reduction interventions
- "But what do the good studies say??" was our question, which meant only RCTs, $\geq$ 25 per arm, direct consumption measure, & $\geq$ 1 day follow-up
- This winnows thousands of papers down to 35 (41

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What was tried?

Table 1. Main findings.

Approach	N (Studies)	N (Estimates)	SMD	τ	95% CIs	p val
Overall	41	112	0.07	0.08	[0.02, 0.12]	0.007
Theory						
Choice Architecture	2	3	0.21	0	[-0.99, 1.42]	0.267
Psychology	9	12	0.11	0.03	[-0.03, 0.24]	0.092
Persuasion	25	77	0.07	0.09	[0.01, 0.13]	0.023
Persuasion & Psychology	11	20	0.11	0.19	[-0.06, 0.28]	0.189
Types of Persuasion						
Animal Welfare	16	65	0.03	0.04	[-0.02, 0.07]	0.189
Environment	16	29	0.09	0.12	[-0.02, 0.19]	0.095
Health	18	29	0.08	0.09	[-0.01, 0.17]	0.064

SMD = standardized mean difference

Main approaches:

- **Persuasion**: animal welfare, environment, health
- **Choice architecture**: nudges, defaults
- **Psychology**: mostly norms experiments

Overall: small, sparse effects — generally changes of a few percentage points. Sad!

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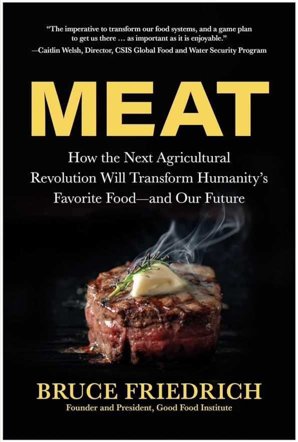
Movement leaders reached the same conclusion



- Around the same time, movement orgs (as far as I can tell) concluded persuasion wasn't working and shifted to
  - **Defaults**: plant-forward menus at scale (e.g., universities, hospitals)
  - **Alt proteins**: Bruce Friedrich and the Good Food Institute
  - **Corporate & political campaigns**: cage-free & BCC commitments, gestation crate bans

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Defaults: sorry, I'm a skeptic

- No defaults study met our inclusion criteria, which means we found no evidence either way on regression to the meat (rebound at the next meal)
- One post-pub study (Ginn & Sparkman 2024) met our criteria and found + results. but also 1) noncompliance at one school 2) lower sales on default days and 3) no measures of spatial spillovers (what people who leave the default station do instead). So net effects on meat are a bit of guesswork (not a knock on the study! Just really hard to do this stuff!)
- Prior: things don't scale up. Especially nduges but really things in general: site selection bias, less skilled/motivated administrators at scale. publication bias, & compensating behavior in the long-run

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Meat alternatives: also a skeptic

- Big question: who's the target audience? Who's going to get on board with cultured meat who isn't already on board with Impossible meat? Can you actually picture her in your mind? How many of her are there out there?
- The meat alternatives industry is really struggling (Beyond Meat down ~97% from peak valuation). What's up with that?
- My belief, and I hope to persuade you about this: the inability of plant-based meats to find their foothold tells you something important about the cultured meat market.
- Survey data suggests a *lot* of resistance (Peacock 2026).

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Recent paper that introduces plant-based alternatives

Food Quality and Preference 143 (2026) 105931

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Food Quality and Preference

journal homepage: www.elsevier.com/locate/foodqual

Effects of adding plant-based menu options on meat selection frequency: A randomized controlled experiment

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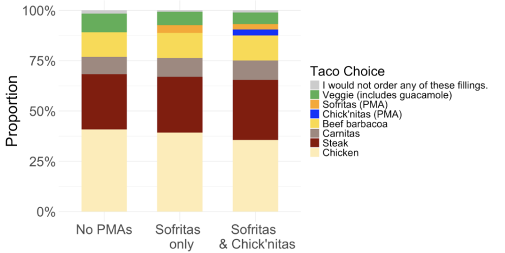
ABSTRACT

Decreasing meat consumption is a critical element of the EAT-Lancet directive to improve human and planetary health, but scalable, effective solutions remain elusive. Plant-based meat analogues are lauded as a promising approach, but their impact on meat demand remains unknown. We tested whether increasing the number of meat analogues on a restaurant menu would decrease meat selection, as well as whether offering a novel chicken-like meat analogue would specifically decrease chicken selection. In a preregistered, randomized, controlled experiment, 4431 English-fluent adults in the U.S. viewed different versions of the menu from Chipotle. Participants in the three arms were shown a Chipotle menu with the pre-existing meat analogue option, "sofritas", removed (0 meat analogues); the standard Chipotle menu (1 meat analogue); or the menu with an added, fictitious, meat analogue, "chick'nitas" (2 meat analogues). Adding one or two meat analogues to the menu did not meaningfully reduce the proportion of participants selecting animal-based meat. Offering one meat analogue versus none produced only a 1.14 percentage point (pp) decrease in meat selection (95% CI [−1.02, 3.30], $P = .30$). For two meat analogues versus none, the estimated decrease was a negligible 2.14 pp. (95% CI [−0.08, 4.36], $P = .06$). However, availability of a chicken meat analogue slightly reduced demand for chicken specifically by −3.65 pp. (95% CI [−7.16, −0.15], $P = .04$). Our findings do not support the hypothesis that expanding meat analogue offerings alone can meaningfully shift consumer choices away from meat.

Figure 3: Taco Filling Choice by Treatment Arm



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Corporate/legal campaigns: I'm fine with "part of the solution"

KFC, Nando's, and others ditch chicken welfare pledge

20 February 2026

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Josh Martin
Business reporter

Major restaurant chains, including KFC, have ditched a commitment to improve chicken sourcing standards in the UK as poultry demand soars.

- Cage-free commitments, BCC, gestation crates, etc.
- But companies can reneg, laws can be overturned.
- A lot of these campaigns are aligned with public opinion, but not with a meaningful voting bloc. Americans oppose gestation crates but obviously DGAF about Save our Bacon Act – not enough to mobilize anyway.
- IMHO we need to move people from "factory farming is bad' to"it's unacceptable."

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So where does that leave us?

- Defaults, alt proteins, corporate campaigns replaced individual persuasion
- I don't see a vegan world emerging from any of these
- Best case: less cruel factory farming in 50 years, but more animals
- But I want something more transformational!!

Let's get weirder

Paul Graham, Black Swan Farming: "The best ideas look initially like bad ideas...If a good idea were obviously good, someone else would already have done it."

- Context here is startup ideas that become huge – bigger than entire alt protein market – but seem ridiculous at the outset, like a business that hinges on people letting strangers into their homes for overnight stays.
- Likewise, defaults, alt proteins, and campaigns allI sound like good ideas. I want to hear more ideas that are like startup ideas: either bad or huge.

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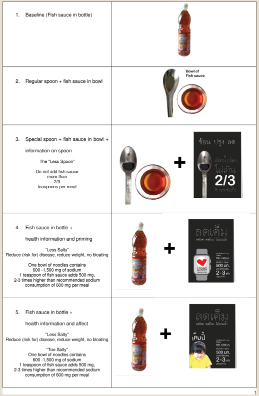
Weird as in “they let you do that??”



1. Baseline (Fish sauce in bottle)	
2. Regular spoon + fish sauce in bowl	Bowl of Fish sauce
3. Special spoon + fish sauce in bowl + information on spoon The “Less Spoon” Do not add fish sauce more than 2/3 teaspoons per meal	+ 2/3
4. Fish sauce in bottle + health information and priming “Less Salty” Reduce (risk for) disease, reduce weight, no bloating One bowl of noodles contains 600 -1,500 mg of sodium 1 teaspoon of fish sauce adds 500 mg, 2-3 times higher than recommended sodium consumption of 600 mg per meal	+
5. Fish sauce in bottle + health information and affect “Less Salty” Reduce (risk for) disease, reduce weight, no bloating “Too Salty” One bowl of noodles contains 600 -1,500 mg of sodium 1 teaspoon of fish sauce adds 500 mg, 2-3 times higher than recommended sodium consumption of 600 mg per meal	+

- Thai cafeterias put holes in fish sauce spoons Kanchanachitra et al. 2020)!
- I think this would never fly in America but who cares? If it works there, great
- Small reduction in fish sauce per person – 0.58g per bowl – but fish sauce fishes are really small so that’s actually a lot of fish lives

Or really not scalable or without any clear application

- You know one cafeteria, one community really well and your idea would only work there
  - "I can't scale this, so why bother?" How about instead let's all do something that works and let's record a lot of qualitative outcomes ("talk to me") so we can build some theory
- Or you might have an idea where you can't see how any advocacy group would take up the mantle
  - Fine with me. Ex: I want to talk to people about fear, early childhood, and food. One on one, really listening. I can't see any way someone would take that baton from me. But it's interesting and it might generate insights and that's plenty ("*Do not trouble your hearts overmuch with thought of the road tonight*").

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So what are your ideas

- Esp. if you want to test them rigorously & want someone to do the IRB paperwork & coauthor the paper, I'm here for it (setgree@stanford.edu)
- So what are your ideas?

P.S. if you want to talk about this later, please email me 1-2 paragraphs describing your study – make sure to mention **Population**, **Intervention** (the thing you're testing), **Comparator (**who's your control group/what are they doing? Is there one?), and **Outcome** (thing you're measuring). I can help you with these if you're not sure yet.

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